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Can Reviewers Catch Bad Agent Patches? A Benchmark Evaluation

Abstract

We built a benchmark of 500 patches produced by autonomous coding agents for real GitHub issues and evaluated an LLM-based review model on it. The review model achieved 81% accuracy overall and is robust to patch size.

1 Method

Each patch is labelled correct if it passes the repository's hidden test suite. We define accuracy as the percentage of patches for which the review model's verdict (approve or reject) matches the test outcome. The review model sees only the issue text and the code diff. We did not measure review time.

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2 Results

Overall accuracy was 81%. Figure 3 breaks accuracy down by patch size (260 small, 180 medium and 60 large patches).

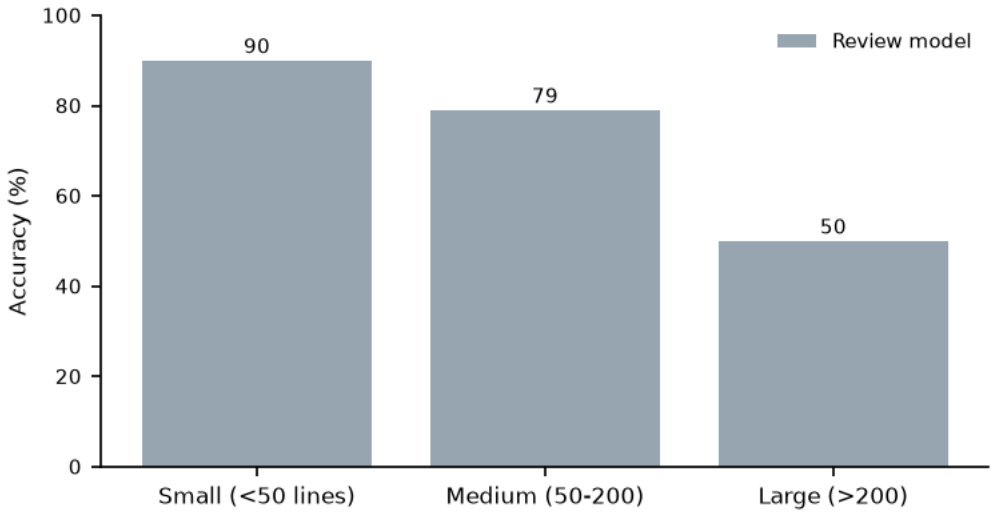


Figure 3: Review-model accuracy (%) by patch size.

3 Discussion

Accuracy remains high across patch sizes, indicating that the review model is robust to patch complexity.

4 Limitations

Only Python repositories were included, and the review model did not have access to the agent's execution traces or test logs.